Machine Learning

25-03-26

features $\longrightarrow$ $\boxed{\text{KNN regression}}$ $\longrightarrow$ target

Hypothesis function

1) get vector representation of features

$V_1 = [a_1 \quad a_2 \quad \dots \quad a_n]$ n - no. of features

2) compute distance of V_1 with all training datapoints

$d_1 \quad d_2 \quad d_3 \dots d_m$ m - no. of training datapoints

3) sort distances in ascending order and find K - nearest neighbors

4) model prediction for V_1 is average of target values of K nearest neighbors

Apply hyp. function on all train and test datapoints

train data

$Y = [Y_1 \quad Y_2 \quad Y_3 \dots Y_m]$ actual target values

$\hat{Y} = [\hat{Y}_1 \quad \hat{Y}_2 \quad \hat{Y}_3 \dots \hat{Y}_m]$ model predicted values

$$MSE = \frac{\sum_{i=1}^{m} (Y_i - \hat{Y}_i)^2}{m}$$ $\longrightarrow$ train mse

test data

$Y = [Y_1 \; Y_2 \; \dots \; Y_p] \rightarrow Y_{actual}$ $\quad$ p - no. of test datapoints

$\hat{Y} = [\hat{Y}_1 \; \hat{Y}_2 \; \dots \; \hat{Y}_p] \rightarrow Y_{pred}$

$$MSE = \frac{\sum_{i=1}^{p} (Y_i - \hat{Y}_i)^2}{p} \rightarrow \text{test mse}$$

If train and test mse both are less then we conclude that model training is successful.

Once the training is successful, we compute accuracy metrics for the model

For regression models, popular metrics are

1) R squared score - R2 score
2) Adjusted R2 score.

$$\text{R2 score} \propto \text{model accuracy} \propto \frac{1}{\text{error}}$$

higher R2 score → lower mse → better model

☆ R2 score

$Y_{act} = Y = [Y_1 \; Y_2 \; Y_3 \dots Y_n]$

$Y_{pred} = \hat{Y} = [\hat{Y}_1 \; \hat{Y}_2 \; \hat{Y}_3 \; \dots \; \hat{Y}_n]$

$$\bar{Y} = \frac{\sum_{i=1}^{n} Y_i}{n} \qquad R2 = 1 - \frac{\sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2}{\sum_{i=1}^{n} (Y_i - \bar{Y})^2}$$

$$R2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2}$$

$$0 \leq R2 \leq 1$$

$0 \rightarrow$ low model accuracy

$1 \rightarrow$ high model accuracy

$R2(m_1) = 0.74$
$R2(m_2) = 0.81$ } m2 is better model

$mse(m_1) = 0.48$
$mse(m_2) = 0.34$ } m2 is better model

Adjusted R2 score

lets say we have trained 4 models

KNN	Linear regr	decision tree regr	neural net regr
4 features	3 features	4 features	5 features

To compare performance of these model, we need to adjust R2 score as they are using different number of features

$$R2_{adj} = 1 - \left(\frac{1 - R2}{n - k - 1}\right)(n-1)$$

R2 → R2 score n - no. of datapoints k - no. of features

lets say $n = 100$

KNN regr with 4 features : $R2 = 0.85$

Linear regr with 3 features : $R2 = 0.85$

$$R2 \text{ adj for KNN} = 1 - \left(\frac{1 - 0.85}{100 - 4 - 1}\right)(100 - 1)$$

$$= 1 - \frac{0.15 \times 99}{95}$$

$$R2 \text{ adj. for Linear regr} = 1 - \frac{0.15 \times 99}{96}$$

R2 adj. for Linear regr > R2 adj for KNN regr

R2 adj score prefers the model delivering higher R2 score with less number of features.

KNN Classification

input $\longrightarrow$ $\boxed{\text{KNN clf}}$ $\longrightarrow$ prediction

$\downarrow$

target is class/category

eg.

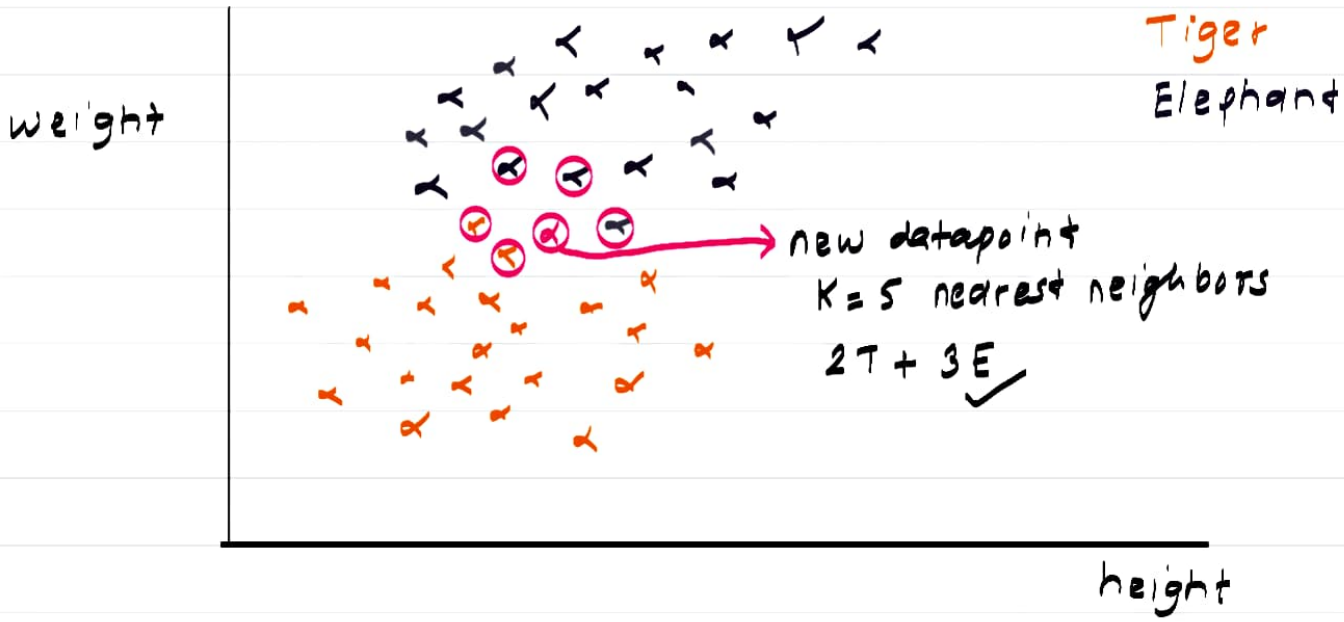
height	weight	class
3.5	50	tiger
5	60	tiger
4	80	elephant
4.2	45	tiger
3.6	62	elephant
⋮	⋮	⋮

height
weight → $\boxed{\text{KNN clf}}$ → T or E

2 classes
binary
classifier
model

binary clf → 2 unique classes are present in target variable

multiclass clf → more than 2 unique classes in target variable



in 5 nearest neighbors 2T and 3 E

KNN clf prediction = mode class of KNN
= E (in this case)

Hypothesis function

- Find K nearest neighbors (using similar steps as that of KNN regression)

- mode of target variable of K nearest neighbors is the model prediction

Performance Evaluation

Regression	Classification
MSE	- simple accuracy
R2 score	- confusion matrix
Adj R2 score	- precision
MAE	- recall
RMSE	- f1 score
	- ROC-AUC

Simple Accuracy

height
weight $\longrightarrow$ $\boxed{\text{KNN clf}}$ $\longrightarrow$ T / E

model is trained

we take 10 datapoints and we get model prediction using their height and weight

Y_{act} = [E T T E T T T E T E]

$\updownarrow$

Y_{pred} = [T T T E E T E E E E]

✗ ✓ ✓ ✓ ✗ ✓ ✗ ✓ ✗ ✓

$$\text{accuracy} = \frac{\text{no. of correct predictions by model}}{\text{total no. of datapoints}}$$

$$\text{accuracy} = \frac{6}{10} = 0.6$$

higher accuracy $\rightarrow$ better model

Accuracy is a good metric in most of the cases, but gives misleading results sometimes

eg.

$Y_{act} = [\ T \quad T \quad T \quad T \quad E \quad E \quad T \quad T \quad T \quad T\ ]$

$Y_{pred} = [\ T \quad T \quad T \quad T \quad T \quad T \quad T \quad T \quad T \quad T\ ]$

$$acc = \frac{8}{10} = 0.8 = 80\%$$

This is case of

<u>class imbalance issue</u>